



दक्षिण बिहार केन्द्रीय विश्वविद्यालय CENTRAL UNIVERSITY OF SOUTH BIHAR

Question Paper of End-Term Examination May 2025

Name of Department	:	Computer Science	Programme	:	M.Sc. (AI)
Session	:	2024-2026	Semester	:	II
Course Title	:	Software Engineering	Course Code	:	CAI82DE01404
Course Credit	:	4	Max. Marks	:	70
			Duration	:	3 hours
Enrolment No. (to be filled by Student)	:	CU 50740233702 8			

Section – A

Answer All the questions in not more than 50 words. Each question carries 1 Points.

(1 x 10 = 10 Points)

1. Define the term Software Engineering.
2. What do we mean by saying that software is invisible?
3. What are the various categories of software?
4. What is software crisis? Was Y2K a software crisis?
5. List out the advantages and drawbacks of waterfall model of software development.
6. What is a Use Case Diagrams?
7. What is a Barry Boehm W⁵HH Principal?
8. What is Delphi Software Cost Estimation Techniques?
9. How black box testing is differing from white box testing?
10. List the problems/challenges facing during maintenance.

Section – B

Answer ANY FIVE questions in not more than 300 words. Each question carries 6 Points.

(6 x 5 = 30 Points)

1. Which life cycle model would you follow for developing software for each of the following applications? Mention the reasons justifying your choice of a particular life cycle model.
 - a) A well-understood data processing system.
 - b) New library automation software that would link various libraries in the city.

[03+03=06]

2. Explain spiral model with a neat diagram.
3. Draw the DFD diagram upto level 3 for university information system.
4. Explain the major responsibilities of a software project manager in details.
5. What is COCOMO? Describe the various modes of COCOMO estimation model.

[02+04=06]

6. Write short notes on any two of the following:
 - a) Types of Couplings
 - b) Boundary value analysis technique
 - c) Reuse oriented software maintenance model

[03+03=06]

Section – C

Answer ANY TWO questions in not more than 1000 words. Each question carries 15 Points.

(15 x 2 = 30 Points)

1. Assume that you are developing an online railway reservation system. Prepare the Software Requirement Specification (SRS) document for the system.
2. Suppose you are the project manager of a software project requiring the following activities.

Activity No.	Activity Name	Duration (weeks)	Immediate Predecessor
1.	Obtain Requirements	4	-
2.	Analyze Operations	4	-
3.	Define subsystems	2	1
4.	Develop database	4	1
5.	Make decision analysis	3	2
6.	Identify constraints	2	5
7.	Build module 1	8	3,4,6
8.	Build module 2	12	3,4,6
9.	Build module 3	18	3,4,6
10.	Write report	10	6
11.	Integration and test	8	7,8,9
12.	Implementation	2	10,11

- (a) Draw the Activity Network representation of the projects.
- (b) Which activities are on the critical path?

[07+08 = 15]

3. Consider a program given below for the classification of a triangle. Its input is a triple of positive integers (say, a, b, c) from interval [1, 100]. The output may be [Scalene, Isosceles, Equilateral, Not a triangle]. Draw and find the following:

- (i) Program flow graph,
- (ii) DD (Decision to Decision) path graph and
- (iii) Find the independent path from the DD path graph.

[05+05+05=15]

```
#include<stdio.h>
#include<conio.h>

1. int main ()
2. {
3.   int a, b, c, validInput=0;
4.   printf("Enter the side 'a' value:");
5.   scanf("%d", &a);
6.   printf("Enter the side 'b' value:");
7.   scanf("%d", &b);
8.   printf("Enter the side 'c' value:");
9.   scanf("%d", &c);
10.  if ((a>0)&&(a<=100) &&
      (b>0)&&(b<=100)&&(c>0)&&(c<=100)){
11.    if((a+b)>c)&&((c+a)>b)&&((c+b)>a)){
12.      validInput=1;
13.    }
14.  }
15.  else{
16.    validInput=-1;
17.  }
18.  if(validInput==1) {
19.    if((a==b)&&(b==c)) {
20.      printf("The triangle is equilateral");
21.    }
22.    elseif ((a==b)||((b==c)||((c==a))) {
23.      printf("The triangle is isosceles");
24.    }
25.    else{
26.      printf("The triangle is scalene");
27.    }
28.  }
29.  else if(validInput==0) {
30.    printf("The value does not constitute
      a Triangle");
31.  }
32.  else {
33.    printf("The input belongs to invalid
      range");
34.  }
35.  getch();
36.  return 1;
37. }
```